

Folie 5 (RC Amplifizierender Schritt Frequenz)

1.

$$RC \frac{d u_o(t)}{dt} + u_o(t) = K u_i(t), \quad K = 1 + \frac{R_2}{R_1}$$

$$u_i(t) = A \sin(\omega t)$$

$$\tau = RC$$

$$K_{VF} = K = 1 + \frac{R_2}{R_1}$$

2.

$$\mathcal{L} \left\{ RC \frac{d u_o(t)}{dt} \right\} + \mathcal{L} \{ u_o(t) \} = \mathcal{L} \{ K u_i(t) \}$$

$$G(s) = \frac{V_o(s)}{V_i(s)}$$

$$\Leftrightarrow RC \mathcal{L} \left\{ \frac{d u_o(t)}{dt} \right\} + V_o(s) = K V_i(s)$$

$$\Leftrightarrow V_o(s) = G(s) V_i(s)$$

$$\Leftrightarrow RC (s V_o(s) - u_o(0)) + V_o(s) = K V_i(s)$$

$$\Leftrightarrow RC s V_o(s) - RC u_o(0) + V_o(s) = K V_i(s)$$

$$\Leftrightarrow RC s V_o(s) + V_o(s) = K V_i(s) + RC u_o(0)$$

$$\Leftrightarrow V_o(s) (RC s + 1) = K V_i(s) + RC u_o(0)$$

$$\Leftrightarrow V_o(s) = \frac{K V_i(s)}{RC s + 1} + \frac{RC u_o(0)}{RC s + 1}$$

$$G(s) = \frac{K}{RC s + 1}$$

3.

$$V_i(s) = \mathcal{L}\{v_i(t)\}$$

$$\Rightarrow V_i(s) = \mathcal{L}\{A \sin(\omega t)\}$$

$$\Rightarrow V_i(s) = A \mathcal{L}\{\sin(\omega t)\}$$

$$\Rightarrow V_i(s) = \frac{A\omega}{s^2 + \omega^2}$$

$$V_o(s) = \frac{KAW}{(s^2 + \omega^2)(\text{Re}s + 1)} + \frac{\text{Re} N_o(0)}{\text{Re}s + 1}$$

$$\Rightarrow N_o(t) = \mathcal{L}^{-1}\left\{ \frac{KAW}{(s^2 + \omega^2)(\text{Re}s + 1)} \right\} + \mathcal{L}^{-1}\left\{ \frac{\text{Re} N_o(0)}{\text{Re}s + 1} \right\}$$

$$\Rightarrow N_o(t) = KAW \mathcal{L}^{-1}\left\{ \frac{1}{(s^2 + \omega^2)(\text{Re}s + 1)} \right\} + \text{Re} N_o(0) \mathcal{L}^{-1}\left\{ \frac{1}{\text{Re}s + 1} \right\}$$

$$\Rightarrow N_o(t) = KAW \mathcal{L}^{-1}\left\{ \frac{1}{\text{Re}(s + \frac{1}{\text{Re}})(s^2 + \omega^2)} \right\} + \frac{\text{Re} N_o(0)}{\text{Re}} \mathcal{L}^{-1}\left\{ \frac{1}{s + \frac{1}{\text{Re}}} \right\}$$

$$\Rightarrow N_o(t) = \frac{KAW}{\text{Re}} \mathcal{L}^{-1}\left\{ \frac{1}{(s + \frac{1}{\text{Re}})(s^2 + \omega^2)} \right\} + N_o(0) e^{-\frac{1}{\text{Re}}t}$$

$$\Rightarrow N_o(t) = \frac{KAW}{\text{Re}} \left(\frac{e^{-\frac{1}{\text{Re}}t}}{(\frac{1}{\text{Re}})^2 + \omega^2} + \frac{\sin(\omega t + \phi)}{\omega \sqrt{(\frac{1}{\text{Re}})^2 + \omega^2}} \right) + N_o(0) e^{-\frac{1}{\text{Re}}t}$$

$$\Rightarrow N_o(t) = \frac{KAW e^{-\frac{1}{\text{Re}}t}}{\text{Re}((\frac{1}{\text{Re}})^2 + \omega^2)} + \frac{KAW \sin(\omega t + \phi)}{\text{Re} \omega \sqrt{(\frac{1}{\text{Re}})^2 + \omega^2}} + N_o(0) e^{-\frac{1}{\text{Re}}t}$$

$$\Rightarrow N_o(t) = \frac{KAW \omega_0 e^{-\frac{\omega_0 t}{\omega_0^2 + \omega^2}}}{\omega_0^2 + \omega^2} + \frac{KAW \sin(\omega t + \phi)}{\sqrt{\omega_0^2 + \omega^2}} + N_o(0) e^{-\frac{\omega_0 t}{\omega_0^2 + \omega^2}}$$

4.

$$N_0(t) = \underbrace{\frac{KA\omega_0 e^{-\omega_0 t}}{\omega_0^2 + \omega^2} + \frac{KA\omega_0 \sin(\omega t + \phi)}{\sqrt{\omega_0^2 + \omega^2}}}_{R. \text{ Forçada}} + \underbrace{N_0(0) e^{-\omega_0 t}}_{R. \text{ Livre}}$$

A resposta forçada tem dois componentes //

5.

$$\lim_{t \rightarrow \infty} N_0(t) \quad \phi = -\arctan(\omega t + \phi) //$$

$$= \lim_{t \rightarrow \infty} \frac{KA\omega_0 e^{-\omega_0 t}}{\omega_0^2 + \omega^2} + \frac{KA\omega_0 \sin(\omega t + \phi)}{\sqrt{\omega_0^2 + \omega^2}} + N_0(0) e^{-\omega_0 t}$$

$$= \lim_{t \rightarrow \infty} \frac{KA\omega_0 \sin(\omega t + \phi)}{\sqrt{\omega_0^2 + \omega^2}}$$

$$= \frac{KA\omega_0}{\sqrt{\omega_0^2 + \omega^2}}$$

$$= \frac{KA\omega_0}{\omega_0^2 (1 + \frac{\omega^2}{\omega_0^2})}$$

$$= \frac{KA\omega_0}{\omega_0 \sqrt{1 + \frac{\omega^2}{\omega_0^2}}}$$

$$= \frac{KA}{\sqrt{1 + (\frac{\omega}{\omega_0})^2}} //$$

6.

6.1.

$$\omega \ll \omega_0 \rightarrow \frac{\omega}{\omega_0} \approx 0$$

$$B = \frac{AK}{\sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}}$$

$$\phi = -\arctan\left(\frac{\omega}{\omega_0}\right)$$

$$\Rightarrow \phi = 0^\circ //$$

$$\Rightarrow B = AK //$$

6.2.

$$\omega \gg \omega_0 \rightarrow \frac{\omega}{\omega_0} \approx \omega$$

$$B = \frac{AK}{\sqrt{1 + \omega^2}}$$

$$\phi = -90^\circ //$$

6.3.

6.3.1. e 6.3.2

$$\omega = \omega_0 \rightarrow \frac{\omega}{\omega_0} = 1$$

$$B = \frac{AK}{\sqrt{1 + 1}}$$

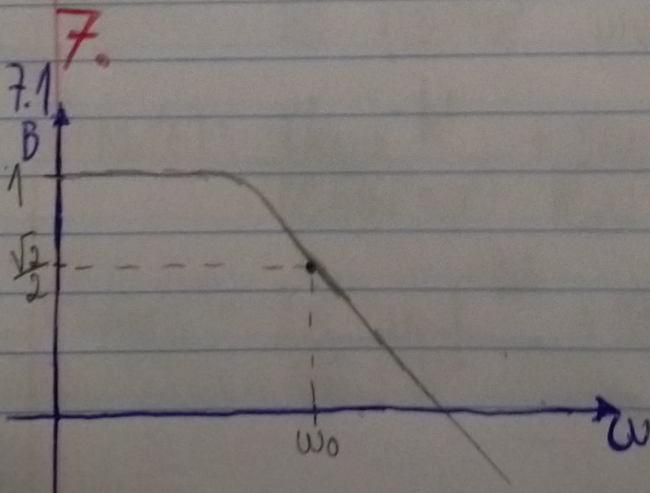
$$\phi = -\arctan(1)$$

$$\Rightarrow \phi = -45^\circ //$$

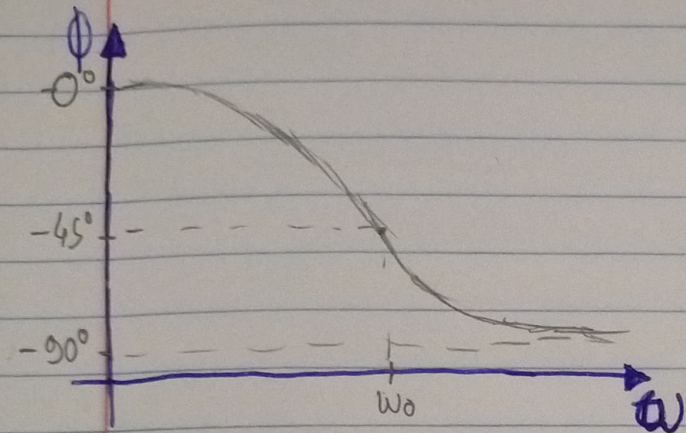
$$\Rightarrow B = \frac{AK}{\sqrt{2}} //$$

$$B = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \rightarrow \omega = \omega_0$$

$$B = AK = 1 \rightarrow \omega \ll \omega_0$$



7.2.



$$\phi = -90^\circ \rightarrow \omega \gg \omega_0$$

$$\phi = -45^\circ \rightarrow \omega = \omega_0$$

$$\phi = 0^\circ \rightarrow \omega \ll \omega_0$$

8.

$$G_{rf} = \frac{1}{\frac{\omega}{\omega_0}} = \frac{\omega_0}{\omega} \downarrow$$

Chama-se filtro passa-baixa pois quando ω é menor G mantém o seu valor, o mesmo não acontece para frequências mais elevadas em que G diminui o seu valor à medida que ω aumenta. //